

Dataset report: PathMNIST histopathology images

Source: Official medmnist.PathMNIST

Dataset **pathmnist** · run 2026-09-29 · job 2836847 · commit 766f1d17e2

Reporting mode: deterministic. This report was rendered from saved, validated evidence by deterministic templates; no language model wrote it. Every number below is read from a checksummed artifact in the evidence bundle.

At a glance

Source	89,996 observations × 2,352 numerical features; the final analysis uses a declared stratified subset (12,500 of 89,996 observations), not the complete dataset
Preprocessing branch	bounded intensity (chosen from the data profile, not the dataset name)
Pilot	1,000 observations; all ten methods; 3 seeds for stochastic methods; 5 refits on 80% subsamples for every method
Retained	PCA, MDS, t-SNE, UMAP
Final cohort	12,500 observations; binding constraint: per-run timeout, final-stage time budget, memory budget at n=15000

Findings

- On the pilot cohort, MDS has the highest pre-registered quality ($Q = 0.666$); no other valid method falls within its equivalence margin.
- Among valid pilot methods, t-SNE has the highest neighbor recall (0.344).
- Among valid pilot methods, MDS has the highest global structure (0.949).
- Among valid pilot methods, Kernel PCA has the highest subsample stability (0.975).
- MDS, t-SNE returned identical embeddings for every seed, so seed stability is trivially 1; the subsample stability used in Q is not affected.
- Across 10 rule variants, PCA, MDS, t-SNE are retained in every variant.
- On the final cohort ($n = 12,500$), t-SNE has the highest trustworthiness (0.900), t-SNE the highest neighbor recall (0.186) and MDS the best global structure (0.955).
- Every final embedding has a negative secondary label silhouette: the annotated classes are not separated in this input representation.
- The first two principal components explain 56.8% of the preprocessed feature variance; 50 components explain 79.2%.

1 · Dataset inspection

Storage	uint8; sparsity 0.0001; values in [0, 255]; non-negative: True; integer-valued: True
Features	0 constant, 0 duplicated; feature variance min/median/max 980.7, 1006, 2133; mitochondrial (MT-) features: 0
Observation totals	row sums min/median/max 1.74e+04, 3.948e+05, 5.742e+05
Memory	dense float64 1615 MB; full pairwise matrix 61793 MB (never materialized by the evaluation code)
Secondary labels	adipose 9,366 · background 9,509 · cancer-associated stroma 9,401 · colorectal adenocarcinoma epithelium 12,885 · debris 10,360 · lymphocytes 10,401 · mucus 8,006 · normal colon mucosa 7,886 · smooth muscle 12,182

Labels only stratify cohort composition and color secondary plots. They never enter preprocessing fits, embeddings or selection.

2 · Preprocessing plan and rationale

Rule evidence: sparsity 0.000, non-negative True, integer-valued True, 2,352 features → branch **bounded intensity**. The same plan is re-fitted inside each cohort (pilot and final).

Step	Parameters (final cohort)	Reason	Fit scope
sample cohort	size=12500; seed=774; stratified=True	Common representative cohort; labels, when present, only preserve class proportions	cohort design
scale intensity	divisor=255	Dense (sparsity 0.000) non-negative integers in [0, 255] behave as 8-bit intensities on one known scale: divide by 255 without per-feature standardization, preserving the pixel-space question	fixed constant
common pca representation	n_components=50	Common denoising and computational input for all ten methods; the displayed PCA is two-dimensional	analysis cohort only

3 • Comparable pilot design

Every method was fitted to the identical saved input (1,000 × 50; SHA-256 c72002bb3331...). Stochastic methods (MDS, GPLVM, t-SNE, UMAP) use 3 seeds. Every method is also refitted on the same 5 random 80% subsamples. Metrics use Euclidean distance in the common input space:

- **T**: trustworthiness at k = 15 (scikit-learn definition).
- **R**: mean k-nearest-neighbor recall, self excluded, stable tie order.
- **G**: Spearman correlation of pairwise distances (all pairs up to 500,000, otherwise a fixed random sample), clipped at 0.
- **S**: mean kNN overlap of subsample refits on their shared observations.

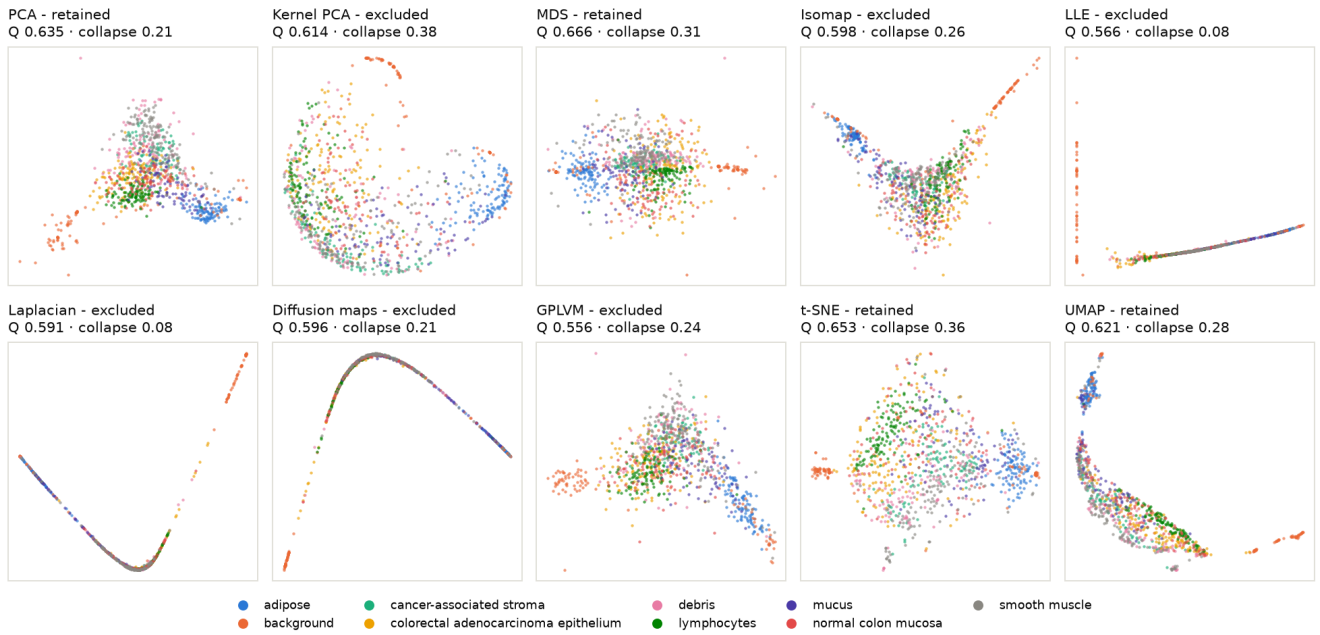
Quality Q = 0.35 trustworthiness + 0.35 neighbor recall + 0.2 global structure + 0.1 stability. The weights and all thresholds were fixed before the real-data run (docs/METHOD_NOTES.md). Runtime is fit_transform time only; imports and JIT warm-up are recorded separately.

Severe diagnostics make a method ineligible: disconnected-graph or singular warnings; a collapsed layout (median collapse ratio < 0.05 and < 0.25 × PCA's); a diffusion Markov spectral gap < 1e-06; SMACOF reaching max_iter.

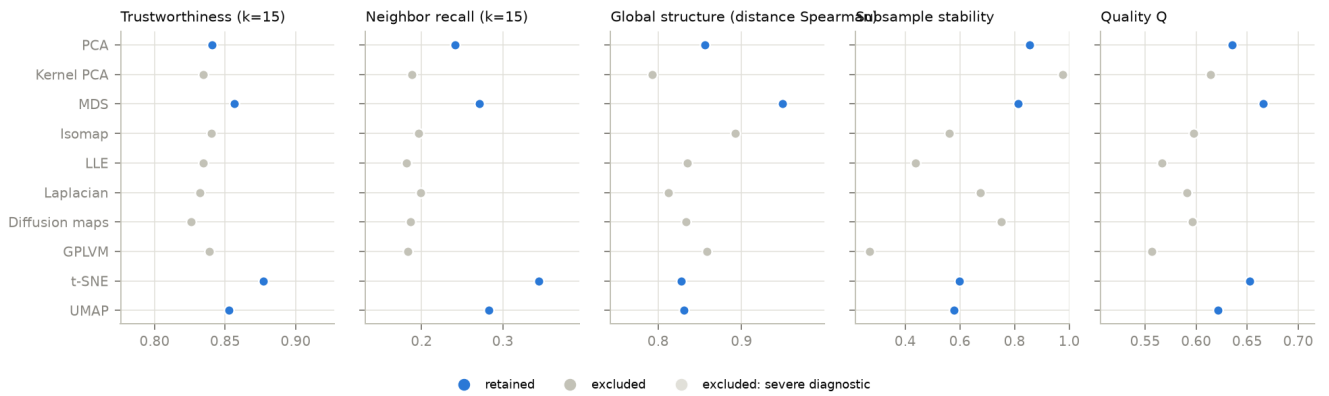
4 • Pilot results

Method	Status	T	R	G	S	Seed S	Q	Fit s	Collapse	Flags
PCA	success	0.841	0.241	0.856	0.856	–	0.635	0.00	0.21	
Kernel PCA	success	0.834	0.188	0.793	0.975	–	0.614	0.04	0.38	
MDS	success	0.857	0.271	0.949	0.813	1.000	0.666	2.32	0.31	seed-invariant
Isomap	success	0.840	0.197	0.892	0.561	–	0.598	0.30	0.26	
LLE	success	0.835	0.181	0.835	0.437	–	0.566	0.21	0.08	
Laplacian	success	0.832	0.199	0.813	0.675	–	0.591	0.09	0.08	
Diffusion maps	success	0.826	0.187	0.833	0.750	–	0.596	0.08	0.21	
GPLVM	success	0.839	0.183	0.859	0.268	0.191	0.556	11.19	0.24	
t-SNE	success	0.877	0.344	0.828	0.598	1.000	0.653	6.19	0.36	seed-invariant
UMAP	success	0.852	0.283	0.831	0.578	0.623	0.621	2.17	0.28	

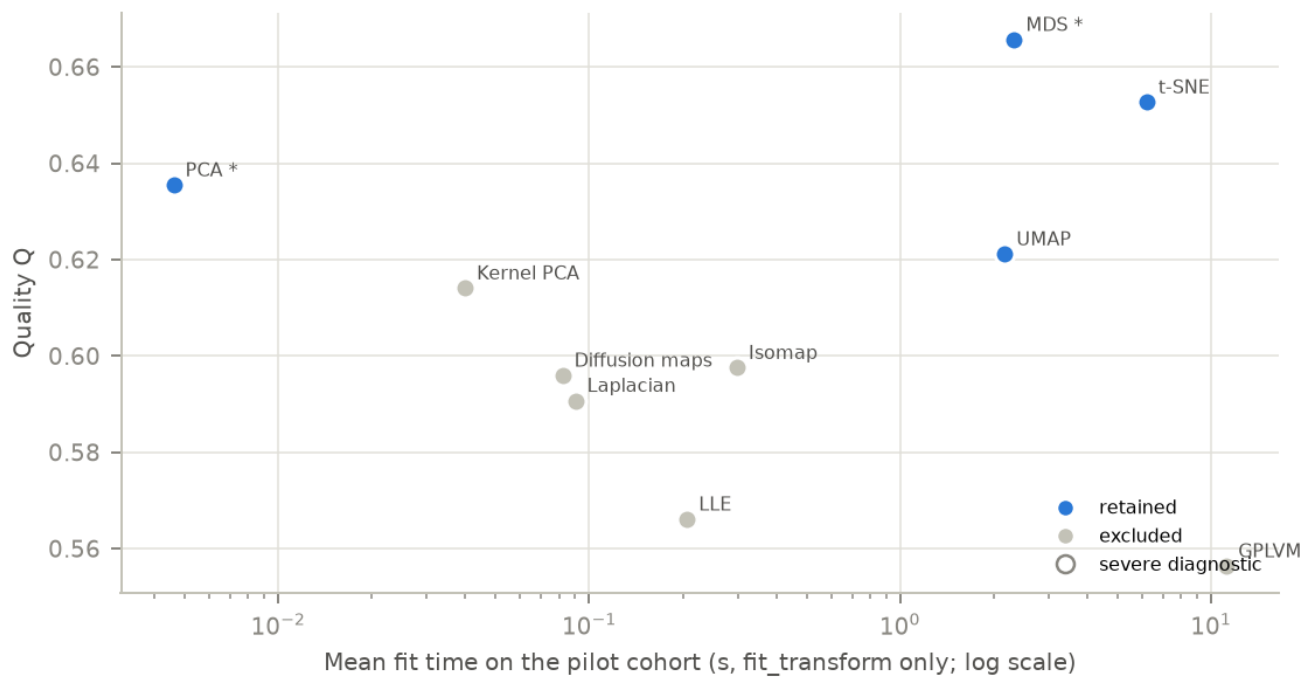
Highlighted rows were retained. Fit s is the mean fit_transform wall time on the full pilot cohort. Seed S is the overlap between repeated seeds on the same cohort; it is shown for stochastic methods only.



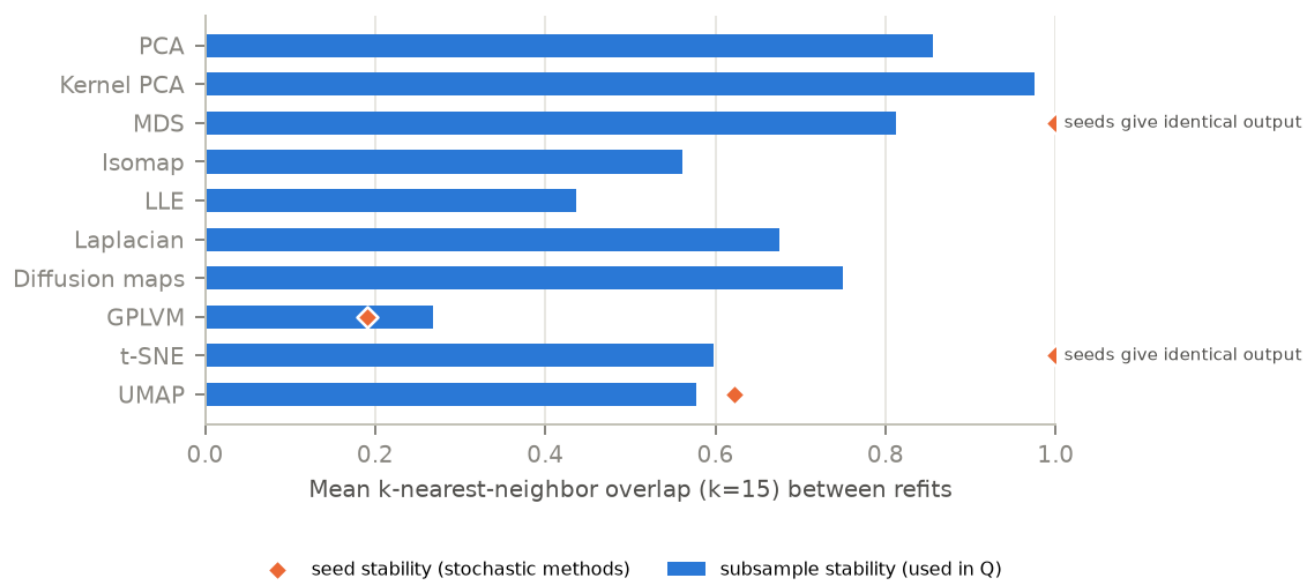
All ten pilot embeddings on the identical input (seed-774 fit). Titles give the pre-registered quality Q, the collapse ratio and the decision; colors are secondary labels only.



Label-free pilot components on the common cohort; each panel's axis spans the observed range. $Q = 0.35 T + 0.35 R + 0.20 G + 0.10 S$.



Fit-only runtime against quality. * marks the epsilon-Pareto frontier (a method is redundant only if another is at least as good and materially better or faster; see the decision rules).



Subsample stability compares refits on overlapping 80% subsamples and applies to every method; seed stability compares repeated seeds on the same cohort.

5 • Evidence-based selection

Best valid pilot quality is 0.6657 (MDS); the 90% threshold is 0.5991.

A method is eligible if it meets that threshold, lies within the practical-equivalence margin of the best, or sits on the ϵ -Pareto frontier. The margin is $\max(2 \times \text{paired subsample SE}, 0.005)$. On the ϵ -Pareto frontier, a runtime difference counts only when it is $\geq 2 \times$ and ≥ 1 s.

PCA is retained as the linear baseline, with at most 4 candidates in total.

Method	Prior (not evidence)	Decision	Measured reason
PCA	Linear variance baseline.	retain	Q=0.6355 vs best mds 0.6657; 90% threshold 0.5991; equivalence margin 0.0050; fit runtime 0.00s. Retained as the linear baseline.
Kernel PCA	Nonlinear kernel structure; quadratic memory.	exclude	Q=0.6141 vs best mds 0.6657; 90% threshold 0.5991; equivalence margin 0.0050; fit runtime 0.04s. Eligible but outside the configured candidate capacity (lower quality rank).

Method	Prior (not evidence)	Decision	Measured reason
MDS	Global pairwise geometry; iterative quadratic distance work.	retain	Q=0.6657 vs best mds 0.6657; 90% threshold 0.5991; equivalence margin 0.0050; fit runtime 2.32s. Retained: meets the 90% threshold, practically equivalent to the best, on the quality-runtime frontier.
Isomap	Geodesic geometry conditional on a connected neighborhood graph.	exclude	Q=0.5975 vs best mds 0.6657; 90% threshold 0.5991; equivalence margin 0.0050; fit runtime 0.30s. Excluded: below threshold, outside the equivalence margin, and made redundant by pca, kernel_pca.
LLE	Local affine reconstruction conditional on well-conditioned neighborhoods.	exclude	Q=0.5661 vs best mds 0.6657; 90% threshold 0.5991; equivalence margin 0.0050; fit runtime 0.21s. Excluded: below threshold, outside the equivalence margin, and made redundant by pca, kernel_pca, isomap.
Laplacian	Local graph geometry conditional on connectivity.	exclude	Q=0.5907 vs best mds 0.6657; 90% threshold 0.5991; equivalence margin 0.0050; fit runtime 0.09s. Excluded: below threshold, outside the equivalence margin, and made redundant by pca, kernel_pca, isomap.
Diffusion maps	Density-corrected diffusion geometry.	exclude	Q=0.5959 vs best mds 0.6657; 90% threshold 0.5991; equivalence margin 0.0050; fit runtime 0.08s. Excluded: below threshold, outside the equivalence margin, and made redundant by pca, kernel_pca.
GPLVM	Probabilistic nonlinear latent-variable model under a fixed optimization budget.	exclude	Q=0.5564 vs best mds 0.6657; 90% threshold 0.5991; equivalence margin 0.0050; fit runtime 11.19s. Excluded: below threshold, outside the equivalence margin, and made redundant by pca, kernel_pca, mds.
t-SNE	Stochastic local-neighborhood visualization.	retain	Q=0.6527 vs best mds 0.6657; 90% threshold 0.5991; equivalence margin 0.0050; fit runtime 6.19s. Retained: meets the 90% threshold.
UMAP	Stochastic neighborhood-graph embedding.	retain	Q=0.6211 vs best mds 0.6657; 90% threshold 0.5991; equivalence margin 0.0050; fit runtime 2.17s. Retained: meets the 90% threshold.

Robustness of the shortlist

Each row below re-runs the same selection mechanics under one rule change on the same saved evidence; only the first row determined the analysis.

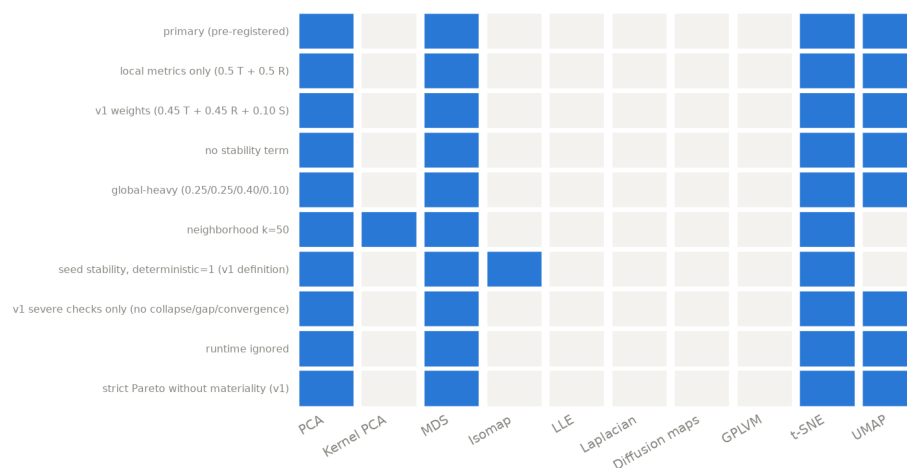
Rule variant	Shortlist
primary (pre-registered)	PCA, MDS, t-SNE, UMAP
local metrics only (0.5 T + 0.5 R)	PCA, t-SNE, UMAP, MDS
v1 weights (0.45 T + 0.45 R + 0.10 S)	PCA, t-SNE, MDS, UMAP
no stability term	PCA, t-SNE, MDS, UMAP
global-heavy (0.25/0.25/0.40/0.10)	PCA, MDS, t-SNE, UMAP
neighborhood k=50	PCA, MDS, t-SNE, Kernel PCA
seed stability, deterministic=1 (v1 definition)	PCA, t-SNE, MDS, Isomap
v1 severe checks only (no collapse/gap/convergence)	PCA, MDS, t-SNE, UMAP
runtime ignored	PCA, MDS, t-SNE, UMAP
strict Pareto without materiality (v1)	PCA, MDS, t-SNE, UMAP

Selection frequency across variants: PCA 100% · Kernel PCA 10% · MDS 100% · Isomap 10% · t-SNE 100% · UMAP 80%.

From the v1 rules to the pre-registered v2 rules

This cumulative ablation is computed on the same evidence; each step changes one rule.

Step	Shortlist
A0 v1 rules on v2 evidence (fit-only timing)	PCA, t-SNE, MDS
A1 + runtime materiality (epsilon-Pareto)	PCA, t-SNE, MDS
A2 + collapse / spectral-gap / convergence checks	PCA, t-SNE, MDS
A3 + global-structure component	PCA, t-SNE, MDS, Isomap
A4 + subsample stability replaces seed stability	PCA, MDS, t-SNE, UMAP
A5 + equivalence-to-best eligibility (= primary)	PCA, MDS, t-SNE, UMAP



Shortlist under alternative rule variants on the same saved pilot evidence (filled = retained). Only the first row determined the analysis; the others test robustness.

6 • Shortlist refinement

Only retained methods were tuned. Each grid point was evaluated with the same seeds, subsamples and quality as the pilot. A non-default point replaces the default only if it gains more than the paired equivalence margin.

Method	Grid	Changed parameters	Status	Q	S	Fit s
PCA	0	default	success	0.635	0.856	0.00
MDS	0	default	success	0.666	0.813	2.32
MDS	1	init=random; n_init=4	success	0.635	0.604	12.93
t-SNE	0	default	success	0.653	0.598	6.24
t-SNE	1	perplexity=15	success	0.640	0.539	5.25
t-SNE	2	perplexity=50	success	0.655	0.619	6.88
UMAP	0	default	success	0.621	0.578	2.15
UMAP	1	n_neighbors=30	success	0.618	0.582	2.50
UMAP	2	n_neighbors=50	success	0.613	0.567	2.71
UMAP	3	min_dist=0.5	success	0.615	0.594	2.16
UMAP	4	min_dist=0.5; n_neighbors=30	success	0.612	0.591	2.46
UMAP	5	min_dist=0.5; n_neighbors=50	success	0.609	0.580	2.73

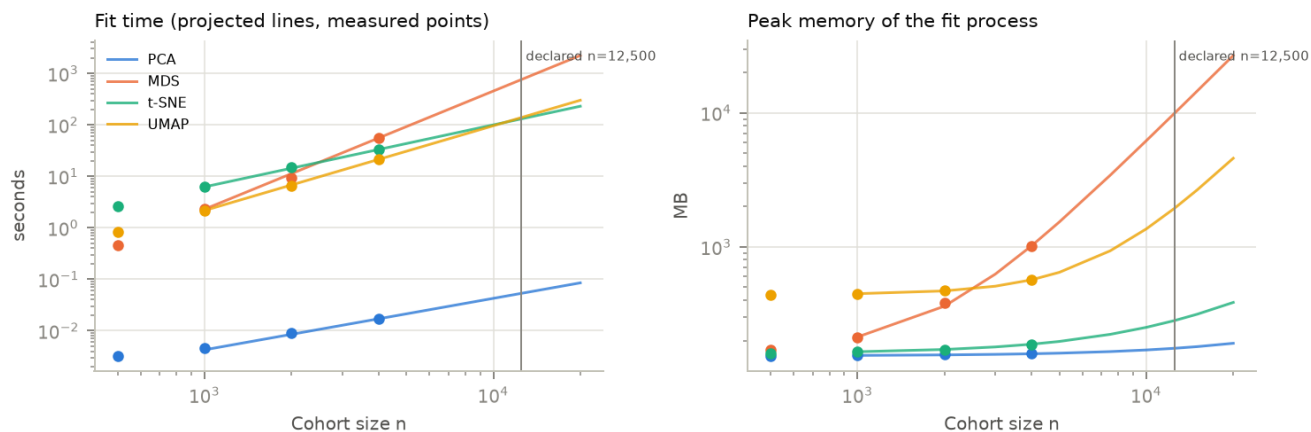
- **PCA**: default has the highest quality.
- **MDS**: default has the highest quality.
- **t-SNE**: best grid point 2 gains 0.0028 <= margin 0.0050; default kept.
- **UMAP**: default has the highest quality.

7 • Final cohort size

Shortlisted methods were fitted on stratified probe cohorts of 500, 1000, 2000, 4000 observations. Log-log slopes (floored at 1) projected their cost at each candidate size, with a 2× safety factor. The limits were: per-run timeout 30 min, final-stage budget 40 min, memory 24 GB.

Declared final size: 12,500 of 89,996 available observations (cap 20,000); binding constraint: per-run timeout, final-stage time budget, memory budget at n=15000.

Method	Time exponent	Memory exponent	Measured at n=4,000	Projected at declared n
MDS	2.30	2.14	55.34 s · 1016 MB	757.3 s · 9897 MB
PCA	0.95	1.18	0.02 s · 159 MB	0.1 s · 175 MB
t-SNE	1.20	1.31	33.11 s · 188 MB	130.6 s · 282 MB
UMAP	1.65	2.17	21.24 s · 567 MB	138.6 s · 1935 MB



Scaling probe of the shortlisted methods and the conservative projections used to declare the final cohort size (per-run timeout, final-stage time budget, memory budget at $n=15000$).

8 · Final results

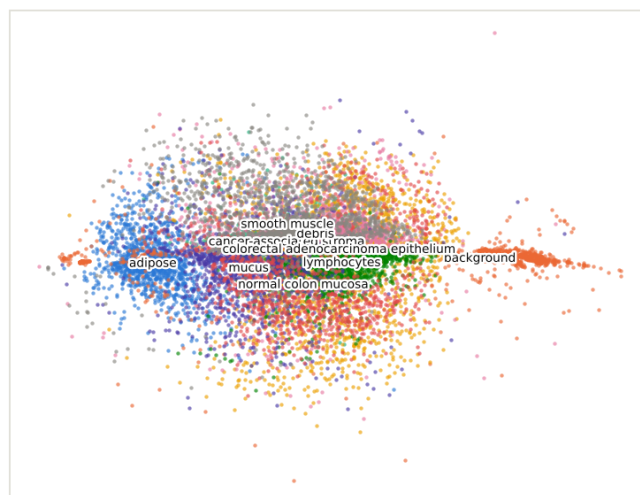
Method	T	R	G	T (k=50)	R (k=50)	Collapse	Silhouette (secondary)	Fit s
PCA	0.843	0.071	0.858	0.844	0.130	0.207	-0.066	0.05
MDS	0.850	0.073	0.955	0.854	0.139	0.303	-0.061	1038.50
t-SNE	0.900	0.186	0.780	0.881	0.201	0.401	-0.066	152.26
UMAP	0.861	0.099	0.831	0.860	0.155	0.246	-0.069	38.48

Final metrics are in-cohort descriptions of the declared final cohort. They are not held-out estimates and should not be compared with pilot values as if the observation sets were identical. Chosen parameters: PCA (svd_solver=full); MDS (init=classical_mds; max_iter=1000; metric=euclidean; metric_mds=True; n_init=1; n_jobs=1; normalized_stress=auto); t-SNE (init=pca; learning_rate=auto; max_iter=750; n_jobs=1; perplexity=30); UMAP (min_dist=0.1; n_epochs=300; n_jobs=1; n_neighbors=15).

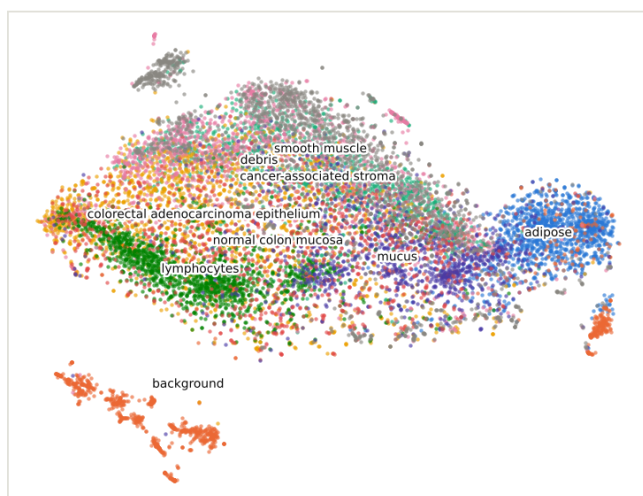
PCA T 0.843 · R 0.071 · G 0.858



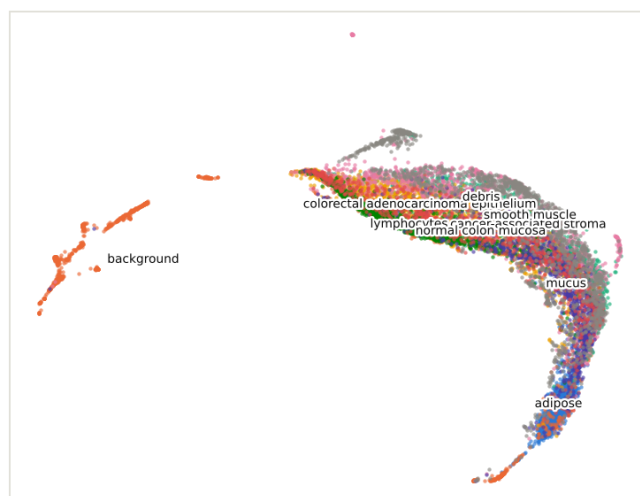
MDS T 0.850 · R 0.073 · G 0.955



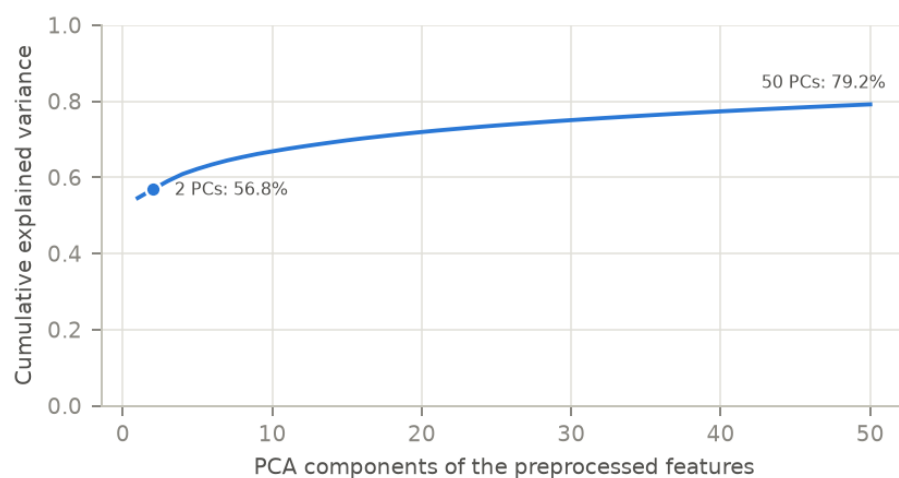
t-SNE T 0.900 · R 0.186 · G 0.780



UMAP T 0.861 · R 0.099 · G 0.831



Final embeddings of the 4 shortlisted methods on the declared final cohort (n=12,500). Class names sit at class medians; colors are secondary labels, and visual separation is not evidence of biological or histological truth.



Variance of the preprocessed feature space retained by the common PCA representation (final cohort).

9 • Limitations

- Q is one pre-registered weighting of label-free criteria. The robustness table shows which conclusions survive other reasonable weightings; no weighting establishes a universally superior method.

- The collapse and spectral-gap checks are conservative and catch only extreme failures. A layout can still be visually degenerate above the thresholds.
- Runtimes are single-threaded CPU measurements on one node, and material-difference tolerances absorb run-to-run noise. The size projections are conservative planning envelopes, not fitted complexity laws.
- GPLVM uses a fixed epoch budget (loss plateau recorded), not a convergence proof.
- PBMC annotations come from a prior clustering, and PathMNIST pixel distances need not reflect histological semantics. Class colors and silhouettes are secondary diagnostics, not validation.

10 • Reproducibility

Configuration SHA-256 7511b092ec09dc3ccb9dd07b022a01b6e98c22724569b54de0c62ec95ecc41af · code SHA-256 c9dc56a74a94fb64c58f5b4e318e267e9ff66183c325f51e8558b23aaf0dd1 · git commit 766f1d17e25a6cd1bdec4dbf62c61140a32e1d3c · job 2836847 on c0402.ll.unc.edu · 2026-09-29T02:00:45.322232+00:00.

Software: numpy 2.4.6 · pandas 2.3.3 · scipy 1.17.1 · scikit-learn 1.9.1 · anndata 0.12.19 · scanpy 1.11.5 · umap-learn 0.5.12 · pydiffmap 0.2.0.1 · medmnist 3.0.2 · torch 2.14.0+cpu · gpytorch 1.15.2 · matplotlib 3.11.2 · weasyprint 70.0 · pypdf 6.19.0 · openai 3.18.0.

Source: <https://zenodo.org/records/10519652/files/pathmnist.npz?download=1> · license CC BY 4.0 · accessed 2026-09-29. Cached file checksums: pathmnist.npz 81823f52dc62...

References: Scanpy PBMC3k documentation and 10x Genomics PBMC3k; Yang et al., MedMNIST v2, Scientific Data 10:41 (2023); Titsias & Lawrence, Bayesian GPLVM, AISTATS (2010); van der Maaten & Hinton, t-SNE, JMLR (2008); McInnes et al., UMAP (2018); Coifman & Lafon, Diffusion maps (2006); Venna & Kaski, trustworthiness (2001). Full links: docs/METHOD_NOTES.md and docs/DATA_PROVENANCE.md.